

A dimension-gap obstruction to non-Hilbertian tangents

FA/Banach run packet for arXiv:2111.06777

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Status. Candidate substantial partial result, likely valid. It proves that a non-Hilbertian tangent at a generic n -dimensional chart point must have Hausdorff dimension at least $n + 1$. The unrestricted question remains open.

Source question and exact scope

The source is D. Lučić, E. Pasqualetto, and T. Rajala, *Non-Hilbertian tangents to Hilbertian spaces*, arXiv:2111.06777v1 [1]. Question 1.5 on PDF page 4 asks:

Is it true that if (X, d, m) is an infinitesimally Hilbertian PI space, then $\text{Tan}_x(X, d, m)$ contains only infinitesimally Hilbertian elements for m -a.e. $x \in X$?

The same page identifies the difficulty. At a generic point of an n -dimensional differentiability chart, each tangent Y has a blow-up map $F : Y \rightarrow T_x X$ that is a metric submersion; although $T_x X$ is Hilbert, Y might have additional independent directions.

Theorem 1.4. *Let $(X, d, \mathfrak{m})$ be an infinitesimally Hilbertian, $\mathfrak{m}$ -rectifiable PI space. Then for $\mathfrak{m}$ -a.e. point $x \in X$ the unique element of $\text{Tan}_x(X, d, \mathfrak{m})$ is infinitesimally Hilbertian.*

Section 3 will be devoted to the proof of Theorem 1.4, which is in fact only a combination of several results that were already available in the literature.

One might wonder whether the $\mathfrak{m}$ -rectifiability assumption in Theorem 1.4 can be dropped. In other words, a natural question is the following:

Question 1.5. Is it true that if $(X, d, \mathfrak{m})$ is an infinitesimally Hilbertian PI space, then $\text{Tan}_x(X, d, \mathfrak{m})$ contains only infinitesimally Hilbertian elements for $\mathfrak{m}$ -a.e. $x \in X$?

We are currently unable to address this question, thus we leave it as an open problem. We point out that a key result on PI spaces, concerning the relation between the (analytic) differential structure and the (geometric) pmGH-tangents, is [8, Theorem 1.12] (see also the related earlier results in [7]). This result says, roughly, that for $\mathfrak{m}$ -a.e. point x of a PI space $(X, d, \mathfrak{m})$ and for every pmGH-tangent $(Y, d_Y, \mathfrak{m}_Y, q) \in \text{Tan}_x(X, d, \mathfrak{m})$, one can construct a pointed blow-up map $\hat{\varphi}: Y \rightarrow T_x X$ (obtained by rescaling a chart φ) which is a metric submersion. Here, $T_x X$ stands for a fiber of the tangent bundle TX , in the sense of Cheeger [7]. In the case where $(X, d, \mathfrak{m})$ is infinitesimally Hilbertian, we have that $T_x X$ is Hilbert for $\mathfrak{m}$ -a.e. $x \in X$, see [14, Section 2.5(3)]. Nevertheless, since $\hat{\varphi}: Y \rightarrow T_x X$ is only a (possibly non-injective) metric submersion, there might be more independent directions on the tangent Y and thus we cannot deduce from this information that the space $(Y, d_Y, \mathfrak{m}_Y)$ is infinitesimally Hilbertian. It is an open problem whether the tangent spaces can really have more independent directions on positively many points. A negative answer to this would resolve Question 1.5 in the affirmative.

We conclude by mentioning that similar results hold also if one considers pmGH-asymptotic cones instead of pmGH-tangent cones. We commit the discussion on the relation between infinitesimal Hilbertianity and asymptotic cones to Appendix A.

Proof intuition

The exact, constant-one nature of the metric submersion retains more information than mere surjectivity. If ℓ is a linear functional on $T_x X$, then $\ell \circ F$ has pointwise Lipschitz constant exactly $\|\ell\|$: every target direction can be lifted without increasing its length. PI calculus therefore turns pullback by F into an isometric embedding of the n -dimensional Hilbert cotangent $T_x^* X$ into the cotangent of Y .

An r -dimensional differentiability chart cannot live in a space of Hausdorff dimension below r . Hence, if $\dim_{\text{H}} Y < n + 1$, the cotangent of Y has rank at most n . The n directions already supplied by F must span it. The entire cotangent norm is then Hilbert, so the Cheeger energy of Y is quadratic. What remains open is exactly the case in which a tangent has room for at least one additional analytic direction.

Dimension-gap theorem

We use the usual Cheeger measurable cotangent bundle of a PI space. The norm of du agrees almost everywhere with the pointwise Lipschitz constant of a Lipschitz function u [2]. An infinitesimally Hilbertian PI space has Hilbertian tangent and cotangent fibers almost everywhere; see the source and the references there.

Lemma 1 (metric-submersion pullback). *Let $F : (Y, d_Y) \rightarrow (V, \|\cdot\|_V)$ be a metric submersion onto a finite-dimensional normed space, and suppose Y is PI. Then, almost everywhere on Y ,*

$$F^* : V^* \longrightarrow T^*Y, \quad \ell \longmapsto d(\ell \circ F),$$

is a linear isometric embedding.

Proof. A metric submersion is 1-Lipschitz and lifts every target point with no increase in distance [2, Definition 1.13]. Therefore, for every $y \in Y$ and $\ell \in V^*$,

$$\text{lip}(\ell \circ F)(y) \leq \|\ell\|_{V^*}.$$

Choose a unit vector $v \in V$ with $\ell(v) = \|\ell\|_{V^*}$. For every $t > 0$, the submersion property gives $y_t \in Y$ such that

$$F(y_t) = F(y) + tv, \quad d_Y(y, y_t) = t.$$

The corresponding difference quotient equals $\|\ell\|_{V^*}$, proving the reverse inequality and hence

$$\text{lip}(\ell \circ F)(y) = \|\ell\|_{V^*} \quad \text{for every } y \in Y. \quad (1)$$

On a PI space, the cotangent norm of a Lipschitz differential agrees almost everywhere with its pointwise Lipschitz constant [2]. Apply this first to a countable dense subset of V^* , intersect the full-measure sets, and then use continuity and linearity. Equation (1) gives $\|F^*\ell\| = \|\ell\|_{V^*}$ for every $\ell \in V^*$ almost everywhere. Thus F^* is a linear isometry, in particular injective. $\square$

Lemma 2 (analytic rank is bounded by Hausdorff dimension). *If a Lipschitz differentiability space Y has an r -dimensional chart of positive measure, then $r \leq \dim_H Y$.*

Proof. For a chart (U, φ) , De Philippis–Marchese–Rindler prove $\varphi_\#(m|_U) \ll \mathcal{L}^r$ [3, Theorem 1.1]. After restricting to a finite positive-measure subset if necessary, this is a nonzero measure, so $\varphi(U)$ has positive r -dimensional Lebesgue measure. Hence $\dim_H \varphi(U) \geq r$. Since φ is Lipschitz, $\dim_H U \geq \dim_H \varphi(U)$, and the claim follows. $\square$

Theorem 1 (one full dimension is necessary). *Let (X, d, m) be an infinitesimally Hilbertian PI space. For m -almost every point x in an n -dimensional differentiability chart, the following holds for every pmGH tangent $(Y, d_Y, m_Y, q) \in \text{Tan}_x(X, d, m)$:*

$$\dim_H Y < n + 1 \implies Y \text{ is infinitesimally Hilbertian.}$$

Equivalently, any infinitesimally non-Hilbertian tangent at such a point has $\dim_H Y \geq n + 1$.

Proof. Choose x in the common full-measure set on which the analytic tangent $V = T_x X$ is an n -dimensional Hilbert space and the blow-up theorem of Cheeger–Kleiner–Schioppa applies to every tangent. For the given Y , it provides a blow-up chart

$$F : Y \longrightarrow V$$

which is a metric submersion [2, Theorem 1.12]; this is precisely the map recalled by the source immediately after Question 1.5.

The space Y is again PI [1], hence it has a measurable cotangent bundle. Lemma 1 supplies n independent cotangent directions at almost every point of Y , so the local analytic rank is at least n . On the other hand, every positive-measure chart has integer dimension at most $\dim_H Y < n + 1$, hence at most n . Thus the cotangent rank is exactly n almost everywhere.

Consequently the isometric embedding $F^* : V^* \rightarrow T_y^* Y$ from Lemma 1 is onto at almost every y . Since V^* is Hilbert, the full pointwise cotangent norm of Y is Hilbert almost everywhere. The Cheeger energy is the integral of the square of this norm, so it satisfies the parallelogram identity. Therefore Y is infinitesimally Hilbertian. $\square$

Corollary 1 (Ahlfors dimension gap). *Suppose in addition that X is Ahlfors Q -regular. At almost every point x of an n -dimensional chart, if $Q < n + 1$, every tangent at x is infinitesimally Hilbertian.*

Proof. Every pmGH tangent is again Ahlfors Q -regular and PI, and therefore has Hausdorff dimension Q . Apply Theorem 1. $\square$

Corollary 2 (the sub-two-dimensional regime). *Every Ahlfors Q -regular PI space with $1 \leq Q < 2$ is infinitesimally Hilbertian, and every one of its pmGH tangents is infinitesimally Hilbertian. Thus Question 1.5 has an affirmative answer in this nonrectifiable regime, even without assuming the original space to be infinitesimally Hilbertian.*

Proof. Every positive-measure differentiability chart has positive integer dimension and, by the preceding rank bound, dimension at most $Q < 2$; hence its dimension is one. Every norm on a one-dimensional vector space is Hilbertian, so the cotangent norm and Cheeger energy are Hilbertian. Tangents preserve the Ahlfors Q -regular PI hypotheses, so the same argument applies to each tangent. $\square$

Limitations, attempts, and novelty check

The theorem does not resolve Question 1.5 when a tangent has Hausdorff dimension at least $n + 1$. In that range, F^*V^* can be a proper Hilbertian subspace of the tangent cotangent, and the metric submersion gives no control on the norm of the additional vertical directions.

Eight focused upgrade attempts were recorded in the accompanying attempt note. They covered direct rank preservation, rank-one inverse limits, products, positive-measure oscillating coefficients, the successful submersion pullback argument, the Hausdorff-dimension closure, the Ahlfors-regular specialization, and a final attempt to remove the dimension gap. The last step runs into exactly the unresolved extra-direction obstruction stated by the source.

Four run indexes and the local parsed corpus were searched for the arXiv id, exact title, Question 1.5, metric submersions, analytic-rank growth, and Ahlfors regular PI spaces. Bounded web/arXiv searches through 17 August 2026 used the exact question/title and combinations of analytic dimension, Hausdorff dimension, metric submersion, $Q < n + 1$, and non-Hilbertian tangent. They found the structural results cited below and adjacent work, but no explicit statement of Theorem 1 and no resolution of the unrestricted question. Novelty confidence is moderate: the argument is a short synthesis of known theorems and may be folklore.

Verification and review recommendation

The proof was audited at the three nonformal interfaces: exact lifting in the metric-submersion lemma, simultaneous almost-everywhere passage from slopes to cotangent norms, and the chart-dimension bound from absolute continuity. No numerical computation is relevant. Recommended human review should focus on whether the chosen PI cotangent formalism identifies the norm of every Lipschitz differential with the pointwise slope exactly (rather than only up to equivalence), and on the standard equivalence between pointwise Hilbertian cotangent norms and quadratic Cheeger energy.

References

- [1] D. Lučić, E. Pasqualetto, and T. Rajala, *Non-Hilbertian tangents to Hilbertian spaces*, Proc. Roy. Soc. Edinburgh Sect. A 153 (2023), 1173–1196; arXiv:2111.06777.
- [2] J. Cheeger, B. Kleiner, and A. Schioppa, *Infinitesimal structure of differentiability spaces, and metric differentiation*, Anal. Geom. Metr. Spaces 4 (2016), 104–159; arXiv:1503.07348.
- [3] G. De Philippis, A. Marchese, and F. Rindler, *On a conjecture of Cheeger*, arXiv:1607.02554 (2016).